# 1 Gain law

(bars denote normalized quantities: lengths by $R$; $a_o := \Delta t/(\gamma_N R)$ [1/pN]; $\bar{a} := a/a_{\mathrm{nom}}$; $\bar{f} := a_o f$, so that $\bar{a}\,\bar{f} = \Delta\bar{w}$; $\bar{w}$ = particle centre along the wall normal, $\bar{w}_s$ = position of the wall surface on the same axis (plane: 0; cell: the visible surface), known, so $\bar{w} - \bar{w}_s = \bar{h}$ with contact at 1; $c_0$, $d_1$ = the [1/1] coefficients, constants of the wall, fitted on the band $1.1 \le \bar{w} - \bar{w}_s \le 5.5$ and used there only; the gain of the form vanishes at $\bar{w} - \bar{w}_s = -c_0$; $e_\theta = \theta - \hat{\theta}$)

$$\bar{a}(\bar{w};\, c_0, d_1) = \frac{(\bar{w} - \bar{w}_s) + c_0}{(\bar{w} - \bar{w}_s) + d_1}$$

$$1 - \bar{a} = \frac{d_1 - c_0}{(\bar{w} - \bar{w}_s) + d_1}$$

$$\bar{a}'(\bar{a};\, c_0, d_1) := \frac{\partial \bar{a}}{\partial \bar{w}} = \frac{d_1 - c_0}{\big((\bar{w} - \bar{w}_s) + d_1\big)^2} = \frac{\big(1 - \bar{a}\big)^2}{d_1 - c_0}$$

$$\Big(\frac{\partial \bar{a}}{\partial c_0}\Big)_{\bar{w},\, d_1} = \frac{1}{(\bar{w} - \bar{w}_s) + d_1} = \frac{1 - \bar{a}}{d_1 - c_0}$$

$$\Big(\frac{\partial \bar{a}}{\partial d_1}\Big)_{\bar{w},\, c_0} = -\,\frac{(\bar{w} - \bar{w}_s) + c_0}{\big((\bar{w} - \bar{w}_s) + d_1\big)^2} = -\,\frac{\bar{a}\,\big(1 - \bar{a}\big)}{d_1 - c_0}$$

$$\frac{\partial \bar{a}'}{\partial \bar{a}} = -\,\frac{2\big(1 - \bar{a}\big)}{d_1 - c_0}, \qquad \frac{\partial \bar{a}'}{\partial d_1} = -\,\frac{\big(1 - \bar{a}\big)^2}{(d_1 - c_0)^2}, \qquad \frac{\partial \bar{a}'}{\partial c_0} = \frac{\big(1 - \bar{a}\big)^2}{(d_1 - c_0)^2}$$

$$a_w = a_o\, \bar{a}_w$$

# 2 True dynamics

$\big(\delta\bar{w}_j[k] := \delta\bar{w}[k - d + j - 1],\ j = 1, \ldots, d+1,\ d = 2\big)$

$$\bar{w}[k] = \bar{w}_d[k] - \delta\bar{w}_3[k]$$

$$\bar{w}_d[k+1] = \bar{w}_d[k] + \Delta\bar{w}_d[k]$$

$$\bar{a}_w[k] = \frac{(\bar{w}[k] - \bar{w}_s) + c_0}{(\bar{w}[k] - \bar{w}_s) + d_1}$$

$$\delta\bar{w}_1[k+1] = \delta\bar{w}_2[k]$$

$$\delta\bar{w}_2[k+1] = \delta\bar{w}_3[k]$$

$$\delta\bar{w}_3[k+1] = \lambda_c\, \delta\bar{w}_3[k] - F_{dw}[k]\, e_{\bar{a}_w}[k] - \varepsilon_w[k]$$

$$\Delta\bar{w}[k] = \Delta\bar{w}_d[k] + (1 - \lambda_c)\, \delta\bar{w}_3[k] + F_{dw}[k]\, e_{\bar{a}_w}[k] + \varepsilon_w[k]$$

$$c_0[k+1] = c_0[k]$$

$$d_1[k+1] = d_1[k]$$

$$F_{dw}[k] = \bar{f}_{dw}[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{f}_{dw}[k-i]$$

$$\varepsilon_w[k] = \bar{a}_w[k]\,\bar{f}_T[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{a}_w[k-i]\,\bar{f}_T[k-i] - (1-\lambda_c)\,n_w[k]$$

## 3 Estimator A: $c_0$ and $d_1$ as states, gain computed (no $a$ slot)

$$\mathbf{x}_A = \left[\, \delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ c_0,\ d_1 \,\right]^\top$$

$$\begin{aligned}
\hat{\bar{w}}[k] &= \bar{w}_d[k] - \delta\hat{\bar{w}}_3[k] \\
\hat{\bar{a}}_w[k] &= \frac{(\hat{\bar{w}}[k] - \bar{w}_s) + \hat{c}_0[k]}{(\hat{\bar{w}}[k] - \bar{w}_s) + \hat{d}_1[k]} \\
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\
\hat{c}_0[k+1] &= \hat{c}_0[k] \\
\hat{d}_1[k+1] &= \hat{d}_1[k]
\end{aligned}$$

## 4 Error dynamics A

$\left(\mathbf{e}_A = [\, e_{\delta\bar{w}_1},\ e_{\delta\bar{w}_2},\ e_{\delta\bar{w}_3},\ e_{c_0},\ e_{d_1} \,]^\top$, coefficients at the estimate$\right)$

$$e_{\bar{a}_w}[k] = -\frac{\left(1-\hat{\bar{a}}_w[k]\right)^2}{\hat{d}_1[k]-\hat{c}_0[k]}\,e_{\delta\bar{w}_3}[k] + \frac{1-\hat{\bar{a}}_w[k]}{\hat{d}_1[k]-\hat{c}_0[k]}\,e_{c_0}[k] - \frac{\hat{\bar{a}}_w[k]\left(1-\hat{\bar{a}}_w[k]\right)}{\hat{d}_1[k]-\hat{c}_0[k]}\,e_{d_1}[k]$$

$$\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \Big(\lambda_c + F_{dw}[k]\,\frac{\left(1-\hat{\bar{a}}_w[k]\right)^2}{\hat{d}_1[k]-\hat{c}_0[k]}\Big)e_{\delta\bar{w}_3}[k] - F_{dw}[k]\,\frac{1-\hat{\bar{a}}_w[k]}{\hat{d}_1[k]-\hat{c}_0[k]}\,e_{c_0}[k] \\
&\quad + F_{dw}[k]\,\frac{\hat{\bar{a}}_w[k]\left(1-\hat{\bar{a}}_w[k]\right)}{\hat{d}_1[k]-\hat{c}_0[k]}\,e_{d_1}[k] - \varepsilon_w[k] \\
e_{c_0}[k+1] &= e_{c_0}[k] \\
e_{d_1}[k+1] &= e_{d_1}[k]
\end{aligned}$$

$$F_e^A[k] = \begin{bmatrix}
0 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 \\
0 & 0 & \lambda_c + F_{dw}[k]\,\frac{(1-\hat{\bar{a}}_w[k])^2}{\hat{d}_1[k]-\hat{c}_0[k]} & -F_{dw}[k]\,\frac{1-\hat{\bar{a}}_w[k]}{\hat{d}_1[k]-\hat{c}_0[k]} & F_{dw}[k]\,\frac{\hat{\bar{a}}_w[k](1-\hat{\bar{a}}_w[k])}{\hat{d}_1[k]-\hat{c}_0[k]} \\
0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 1
\end{bmatrix}$$

## 5 Estimator B: $a$ slot carried along the law, $c_0$ fixed by the seed

$\big(M[k] := \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]$; $c_0$ is not a state, it enters the law's slope through $d_1 - c_0$ only$\big)$

$$\mathbf{x}_B = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ d_1\,\big]^\top$$

$$\begin{aligned}
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\
\hat{\bar{a}}_w[k+1] &= \hat{\bar{a}}_w[k] + \frac{\big(1-\hat{\bar{a}}_w[k]\big)^2}{\hat{d}_1[k]-c_0}\,M[k] \\
\hat{d}_1[k+1] &= \hat{d}_1[k]
\end{aligned}$$

## 6 Error dynamics B

$\big(\mathbf{e}_B = [\,e_{\delta\bar{w}_1},\ e_{\delta\bar{w}_2},\ e_{\delta\bar{w}_3},\ e_{\bar{a}_w},\ e_{d_1}\,]^\top$, coefficients at the estimate$\big)$

$$\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c\,e_{\delta\bar{w}_3}[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= (1-\lambda_c)\,\frac{\big(1-\hat{\bar{a}}_w[k]\big)^2}{\hat{d}_1[k]-c_0}\,e_{\delta\bar{w}_3}[k] \\
&\quad + \Big(1 + \frac{\big(1-\hat{\bar{a}}_w[k]\big)^2}{\hat{d}_1[k]-c_0}\,F_{dw}[k] - \frac{2\big(1-\hat{\bar{a}}_w[k]\big)}{\hat{d}_1[k]-c_0}\,M[k]\Big)e_{\bar{a}_w}[k] \\
&\quad - \frac{\big(1-\hat{\bar{a}}_w[k]\big)^2}{(\hat{d}_1[k]-c_0)^2}\,M[k]\,e_{d_1}[k] + \frac{\big(1-\hat{\bar{a}}_w[k]\big)^2}{\hat{d}_1[k]-c_0}\,\varepsilon_w[k] \\
e_{d_1}[k+1] &= e_{d_1}[k]
\end{aligned}$$

$$F_e^B[k] = \begin{bmatrix}
0 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 \\
0 & 0 & \lambda_c & -F_{dw}[k] & 0 \\
0 & 0 & (1-\lambda_c)\,\frac{(1-\hat{\bar{a}}_w[k])^2}{\hat{d}_1[k]-c_0} & 1 + \frac{(1-\hat{\bar{a}}_w[k])^2}{\hat{d}_1[k]-c_0}\,F_{dw}[k] - \frac{2(1-\hat{\bar{a}}_w[k])}{\hat{d}_1[k]-c_0}\,M[k] & -\frac{(1-\hat{\bar{a}}_w[k])^2}{(\hat{d}_1[k]-c_0)^2}\,M[k] \\
0 & 0 & 0 & 0 & 1
\end{bmatrix}$$

## 7 Measurement

$\big(\nabla_d\bar{w}_d[k] := \bar{w}_d[k] - \bar{w}_d[k-d]$; $\hat{\bar{w}}_d[k] := \hat{\bar{w}}[k] - \nabla_d\bar{w}_d[k]\big)$

$$y_1[k] = \delta\bar{w}_1[k] + n_w[k]$$
$$y_2[k] = \bar{a}_w[k-d] + n_a[k]$$
$$\hat{y}_1[k] = \delta\hat{\bar{w}}_1[k]$$

A:
$$\hat{\bar{a}}_d[k] = \frac{(\hat{\bar{w}}_d[k] - \bar{w}_s) + \hat{c}_0[k]}{(\hat{\bar{w}}_d[k] - \bar{w}_s) + \hat{d}_1[k]}$$
$$\hat{y}_2[k] = \hat{\bar{a}}_d[k]$$
$$H^A[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\dfrac{\big(1-\hat{\bar{a}}_d[k]\big)^2}{\hat{d}_1[k]-\hat{c}_0[k]} & \dfrac{1-\hat{\bar{a}}_d[k]}{\hat{d}_1[k]-\hat{c}_0[k]} & -\dfrac{\hat{\bar{a}}_d[k]\big(1-\hat{\bar{a}}_d[k]\big)}{\hat{d}_1[k]-\hat{c}_0[k]} \end{bmatrix}$$

B:
$$\hat{\bar{a}}_d[k] = \hat{\bar{a}}_w[k] - \frac{\big(1-\hat{\bar{a}}_w[k]\big)^2}{\hat{d}_1[k]-c_0}\,\nabla_d\bar{w}_d[k]$$
$$\hat{y}_2[k] = \hat{\bar{a}}_d[k]$$
$$H^B[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1+\dfrac{2\big(1-\hat{\bar{a}}_w[k]\big)}{\hat{d}_1[k]-c_0}\,\nabla_d\bar{w}_d[k] & \dfrac{\big(1-\hat{\bar{a}}_w[k]\big)^2}{(\hat{d}_1[k]-c_0)^2}\,\nabla_d\bar{w}_d[k] \end{bmatrix}$$

$$\mathbf{e}_y[k] = H[k]\,\mathbf{e}^-[k] + \begin{bmatrix} n_w[k] \\ n_a[k] \end{bmatrix}$$

# 8 Noise

$$Q^A[k] = Q_{33}[k]\,\mathbf{g}_A\,\mathbf{g}_A^\top, \qquad \mathbf{g}_A = \big[\,0,\ 0,\ -1,\ 0,\ 0\,\big]^\top$$
$$Q^B[k] = Q_{33}[k]\,\mathbf{g}_B\,\mathbf{g}_B^\top, \qquad \mathbf{g}_B = \Big[\,0,\ 0,\ -1,\ \frac{\big(1-\hat{\bar{a}}_w[k]\big)^2}{\hat{d}_1[k]-c_0},\ 0\,\Big]^\top$$
$$Q_{33}[k] = \kappa_T\Big(\hat{\bar{a}}_w[k] + (1-\lambda_c)^2\big(\hat{\bar{a}}_w[k-1] + \hat{\bar{a}}_w[k-2]\big)\Big) + (1-\lambda_c)^2\sigma_{n_w}^2, \qquad \kappa_T = \frac{4k_BT\,a_o}{R}$$
$$R[k] = \mathrm{diag}\big(R_1,\ R_2[k]\big), \qquad R_1 = \frac{\sigma_n^2}{R^2}, \qquad R_2[k] = 2a_{\mathrm{cov}}^2\,IF_{\mathrm{var}}\big(\hat{\bar{a}}_w[k]\big)\big(\hat{\bar{a}}_w[k]+\xi\big)^2$$

# 9 Observability

($\Phi(k, k_0) := F_e[k-1] \cdots F_e[k_0]$, $\Phi(k_0, k_0) = I$; $N$ window length)

($O_w := R_w^{-1/2} \, [\, H[k_0]\Phi(k_0, k_0);\ \ldots;\ H[k_0+N]\Phi(k_0+N, k_0)\,]$; $O_w \, \mathbf{v}_i = \sigma_i \, \mathbf{u}_i$, $\sigma_1 \geq \cdots \geq \sigma_5$; $\sigma_\theta := \sqrt{[(O_w^\top O_w)^{-1}]_{\theta\theta}}$)

A: $\dfrac{H^A_{25}[k]}{H^A_{24}[k]} = -\,\hat{\bar{a}}_d[k], \qquad \dfrac{F^A_{e,35}[k]}{F^A_{e,34}[k]} = -\,\hat{\bar{a}}_w[k]$

$\Delta\bar{w}_d[k] = 0 \ \ \forall k, \ \delta\hat{\bar{w}}_3 \to 0 \quad \Longrightarrow \quad F_{dw}[k] \to 0, \ M[k] \to 0, \ \nabla_d \bar{w}_d[k] = 0, \ \hat{\bar{a}}_w[k] = \hat{\bar{a}}_w$ constant

A: $H^A[k]_{2,4:5} \to \dfrac{1 - \hat{\bar{a}}_w}{\hat{d}_1 - \hat{c}_0} \, \big[\, 1, \ -\hat{\bar{a}}_w \,\big]$ constant in $k$

$\Longrightarrow \quad \mathbf{e} \propto [\, 0, \ 0, \ 0, \ \hat{\bar{a}}_w, \ 1 \,]^\top$ unobservable

B: $H^B[k]_{2,4:5} \to \big[\, 1, \ 0 \,\big], \qquad F^B_e[k]_{:,5} \to \big[\, 0, \ 0, \ 0, \ 0, \ 1 \,\big]^\top$

$\Longrightarrow \quad e_{d_1}$ unobservable

# 10 Fixed height

($\bar{w}_d[k] = \bar{w}_h$ for all $k$; $g_h := \bar{w}_h - \bar{w}_s$; $\bar{a}_h := \bar{a}(\bar{w}_h)$ the plant's gain there; $r$ = the one number a single height pins; $\mathbf{u}_h$, $\mathbf{v}_h$ = the unlearnable and the learnable direction in $(c_0, d_1)$)

$$\frac{g_h + c_0}{g_h + d_1} = \bar{a}_h \quad \Longleftrightarrow \quad c_0 - \bar{a}_h \, d_1 = g_h \, (\bar{a}_h - 1)$$

$$r(\hat{c}_0, \hat{d}_1;\, \bar{w}_h) := \hat{c}_0 - \bar{a}_h \, \hat{d}_1 - g_h \, (\bar{a}_h - 1)$$

$$\mathbf{v}_h = \frac{[\, 1, \ -\bar{a}_h \,]^\top}{\sqrt{1 + \bar{a}_h^2}}, \qquad \mathbf{u}_h = \frac{[\, \bar{a}_h, \ 1 \,]^\top}{\sqrt{1 + \bar{a}_h^2}}, \qquad \mathbf{u}_h^\top \mathbf{v}_h = 0$$

A: $H^A[k]_{2,4:5} = \dfrac{1 - \hat{\bar{a}}_d}{\hat{d}_1 - \hat{c}_0} \sqrt{1 + \hat{\bar{a}}_d^2} \, \mathbf{v}_h^\top \qquad (\delta\hat{\bar{w}}_3 = 0)$

two heights $\bar{w}_1 \neq \bar{w}_2$ : $\begin{bmatrix} 1 & -\bar{a}_1 \\ 1 & -\bar{a}_2 \end{bmatrix} \begin{bmatrix} c_0 \\ d_1 \end{bmatrix} = \begin{bmatrix} g_1(\bar{a}_1 - 1) \\ g_2(\bar{a}_2 - 1) \end{bmatrix}$

$$d_1 = \frac{g_1(\bar{a}_1 - 1) - g_2(\bar{a}_2 - 1)}{\bar{a}_2 - \bar{a}_1}, \qquad c_0 = g_1(\bar{a}_1 - 1) + \bar{a}_1 \, d_1$$

B at $\bar{w}_h$ : $F^B_e[k]_{4,5} = -\dfrac{\big(1 - \hat{\bar{a}}_w[k]\big)^2}{(\hat{d}_1[k] - c_0)^2} \, (1 - \lambda_c) \, \delta\hat{\bar{w}}_3[k], \qquad H^B[k]_{2,5} = 0$

## Seed and prior

plane (band fit; not used, the plane keeps its polynomial): $(c_0, d_1) = (-0.984,\ 0.161)$

cell (Fig. 11 band fit): $(c_0, d_1) = (0.056,\ 1.197)$, $\bar{a}_w(\bar{w} - \bar{w}_s = 1.1) = 0.503$

wall family prior (half the plane–cell spread): $\sigma_{c_0,0} = \sigma_{d_1,0} = 0.52$; tight: 0.05

$$\hat{c}_0[0] = c_{0,\text{seed}}, \qquad \hat{d}_1[0] = d_{1,\text{seed}}$$

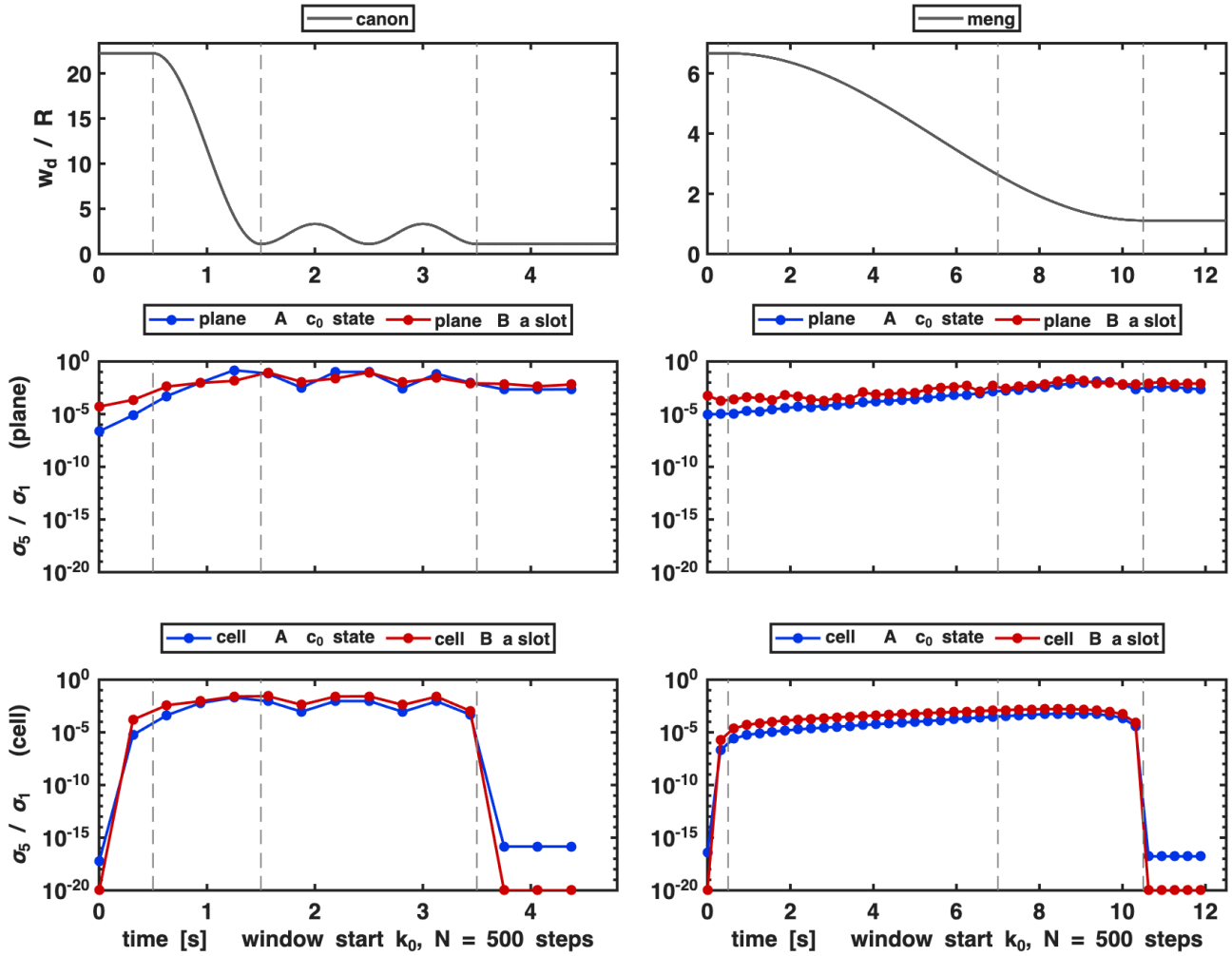
$$\hat{\bar{a}}_w[0] = \frac{(\bar{w}[0] - \bar{w}_s) + \hat{c}_0[0]}{(\bar{w}[0] - \bar{w}_s) + \hat{d}_1[0]}$$

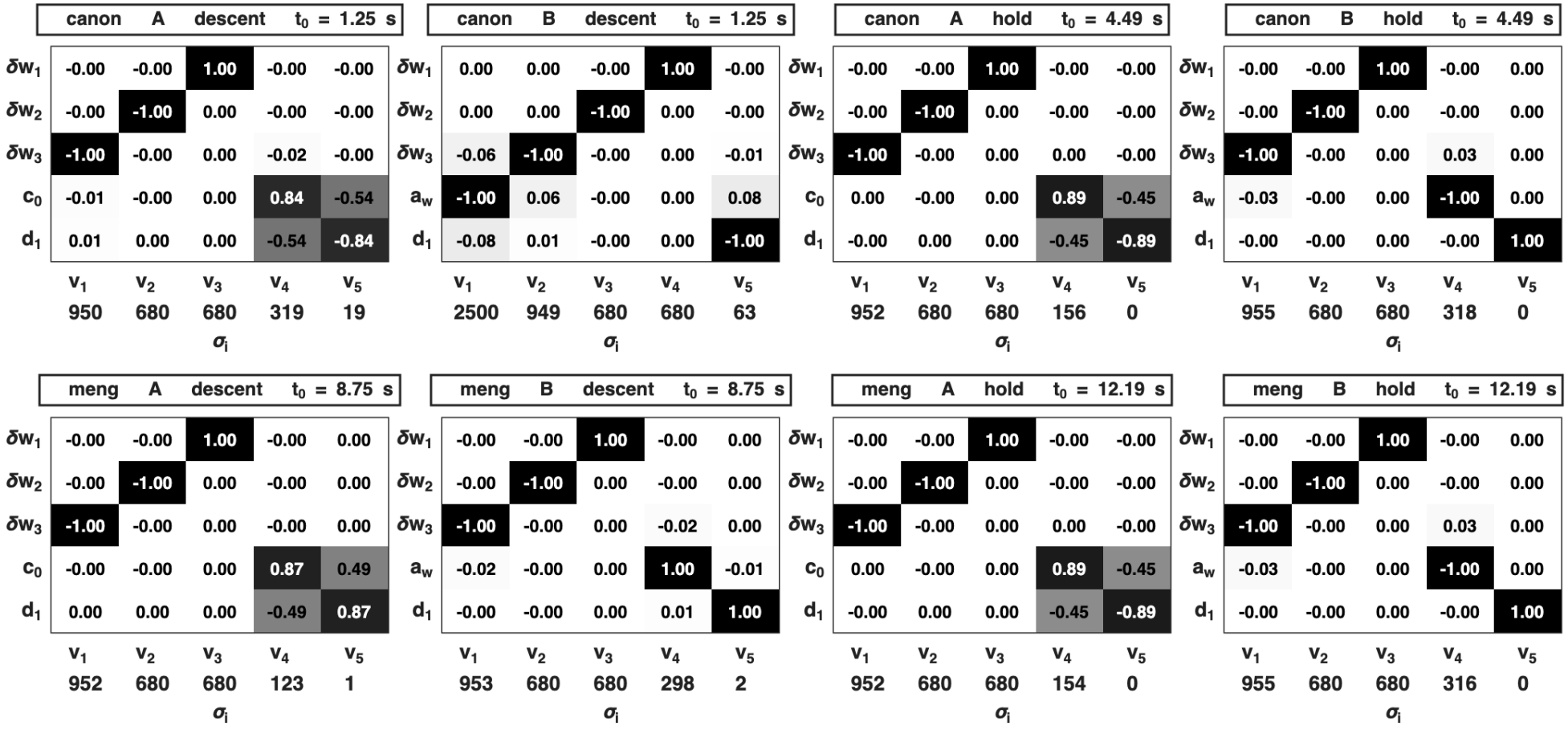
$$P[0] = \operatorname{diag}\big(P_{11}[0],\ P_{22}[0],\ P_{33}[0],\ P_{44}[0],\ P_{55}[0]\big)$$

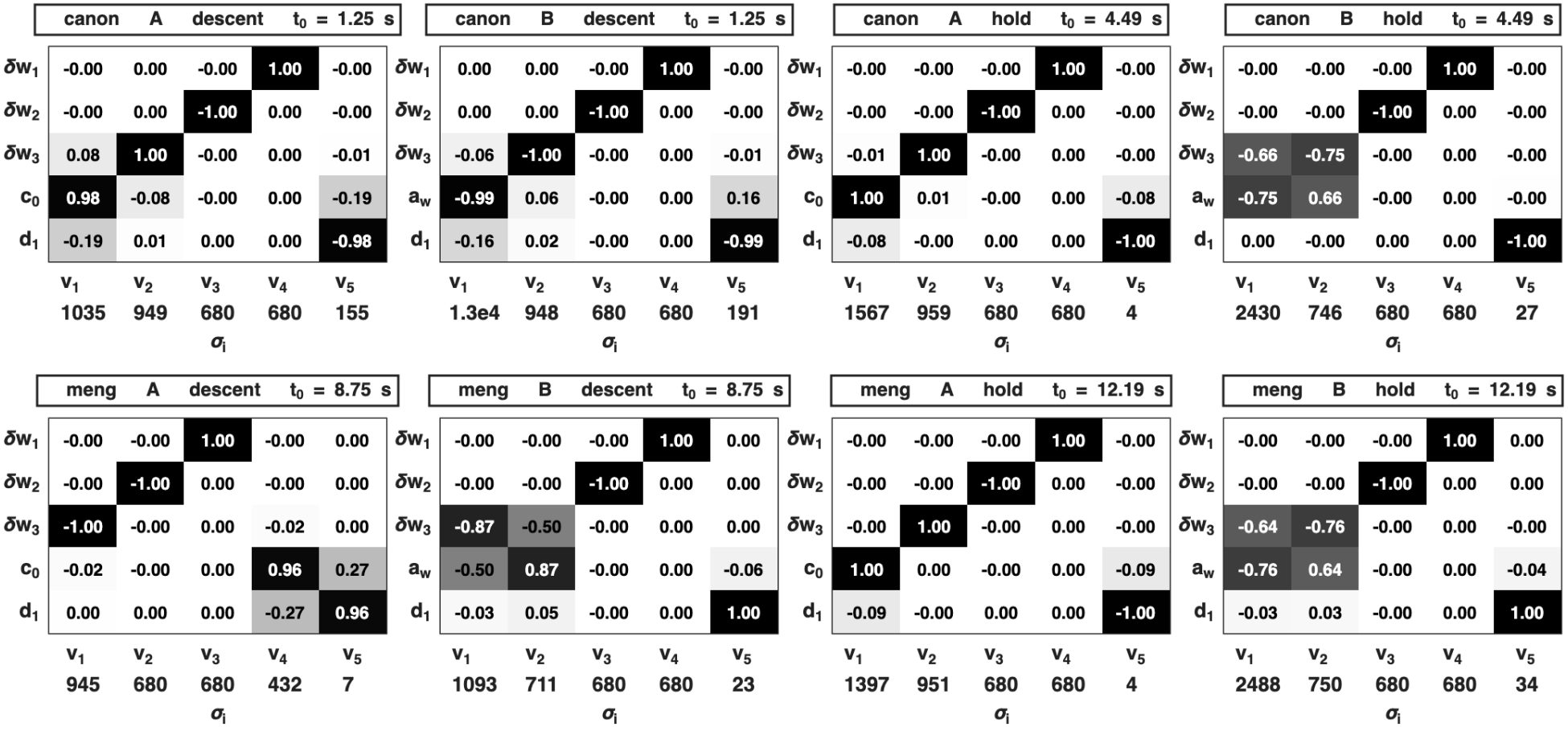
A: $$\sqrt{P_{44}[0]} = \sigma_{c_0,0}, \qquad \sqrt{P_{55}[0]} = \sigma_{d_1,0}$$

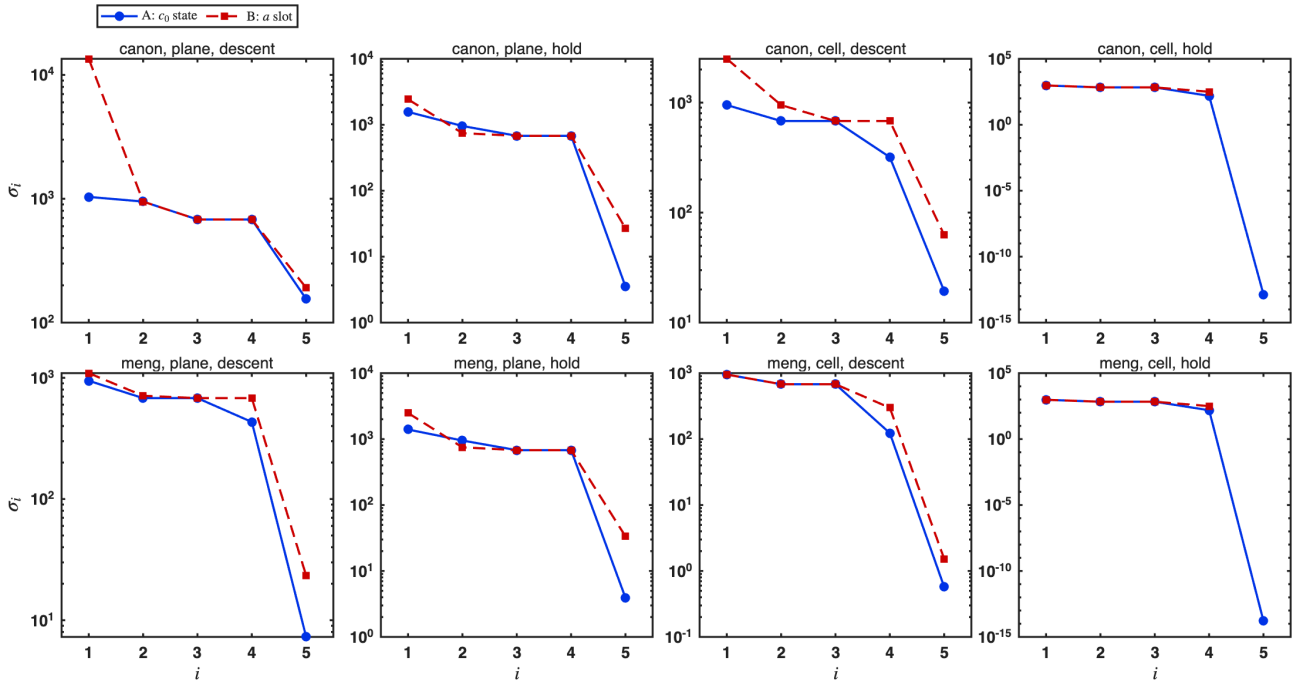
B: $$P_{44}[0] = \Big(\frac{1 - \hat{\bar{a}}_w[0]}{\hat{d}_1[0] - c_0}\Big)^2 \sigma^2_{c_0,0} + \Big(\frac{\hat{\bar{a}}_w[0](1 - \hat{\bar{a}}_w[0])}{\hat{d}_1[0] - c_0}\Big)^2 \sigma^2_{d_1,0}$$

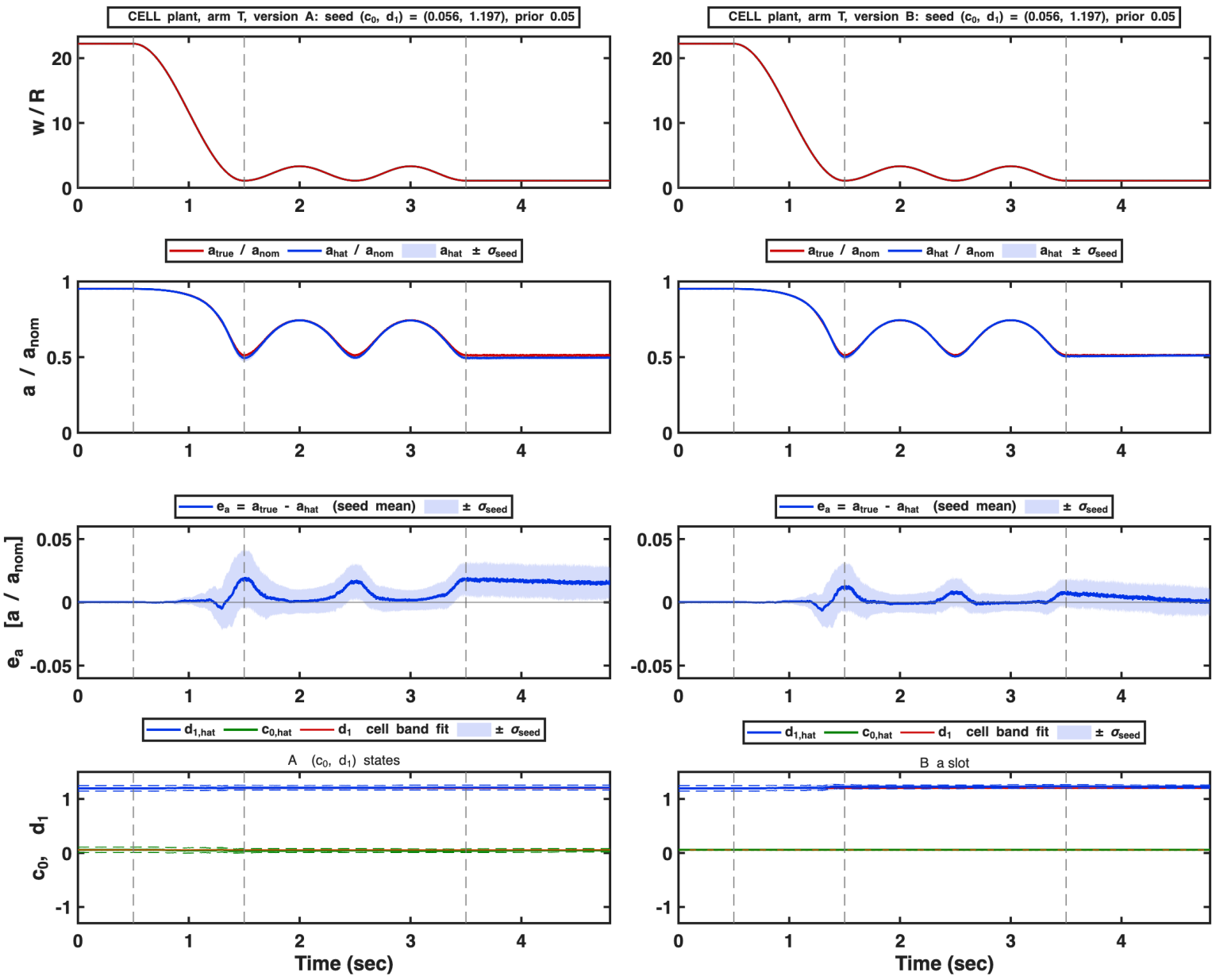
B: $$\sqrt{P_{55}[0]} = \sigma_{d_1,0}$$

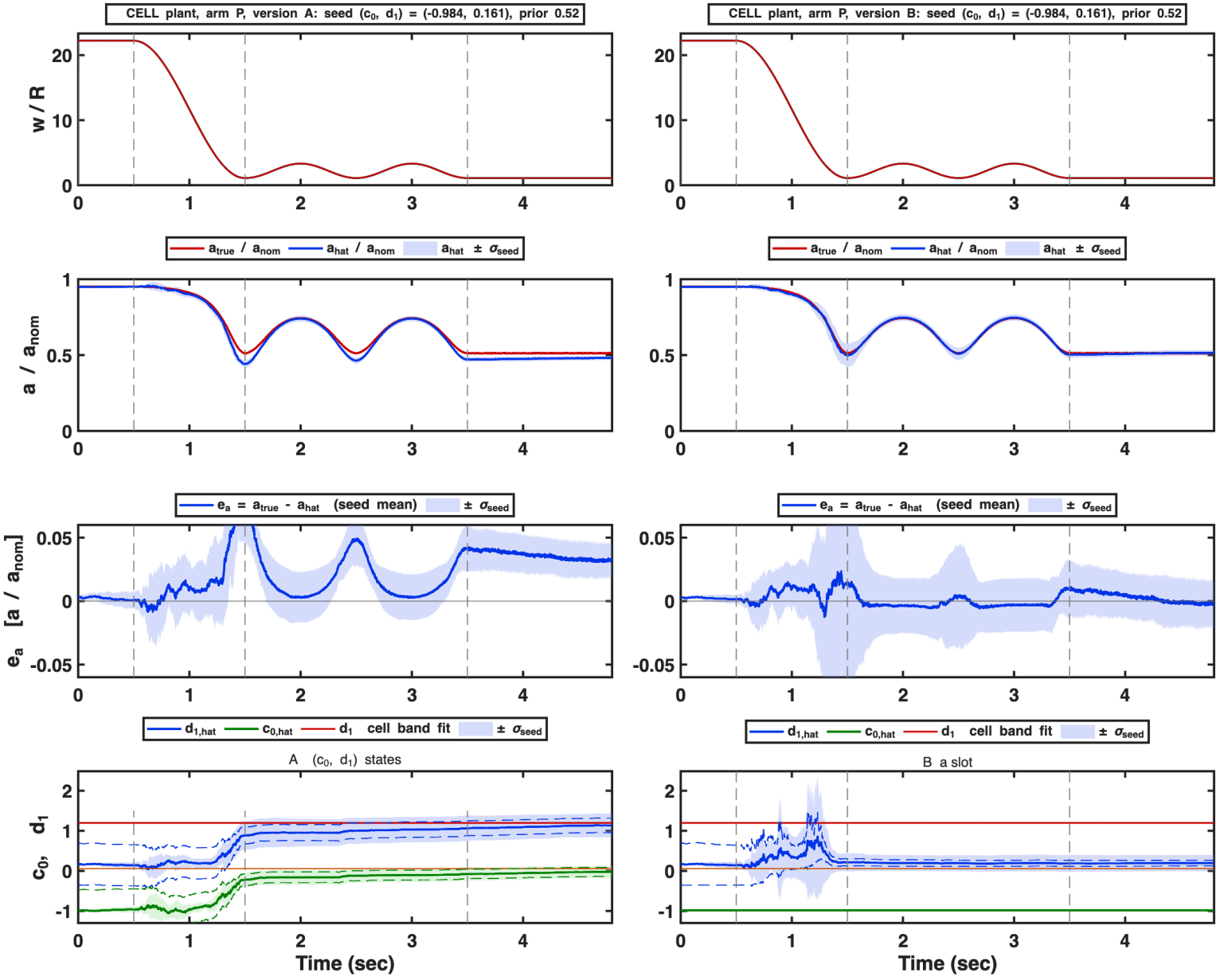

canon
meng
$w_d$ / R
plane A $c_0$ state
plane B a slot
$\sigma_5$ / $\sigma_1$ (plane)
cell A $c_0$ state
cell B a slot
$\sigma_5$ / $\sigma_1$ (cell)
time [s] window start $k_0$, N = 500 steps

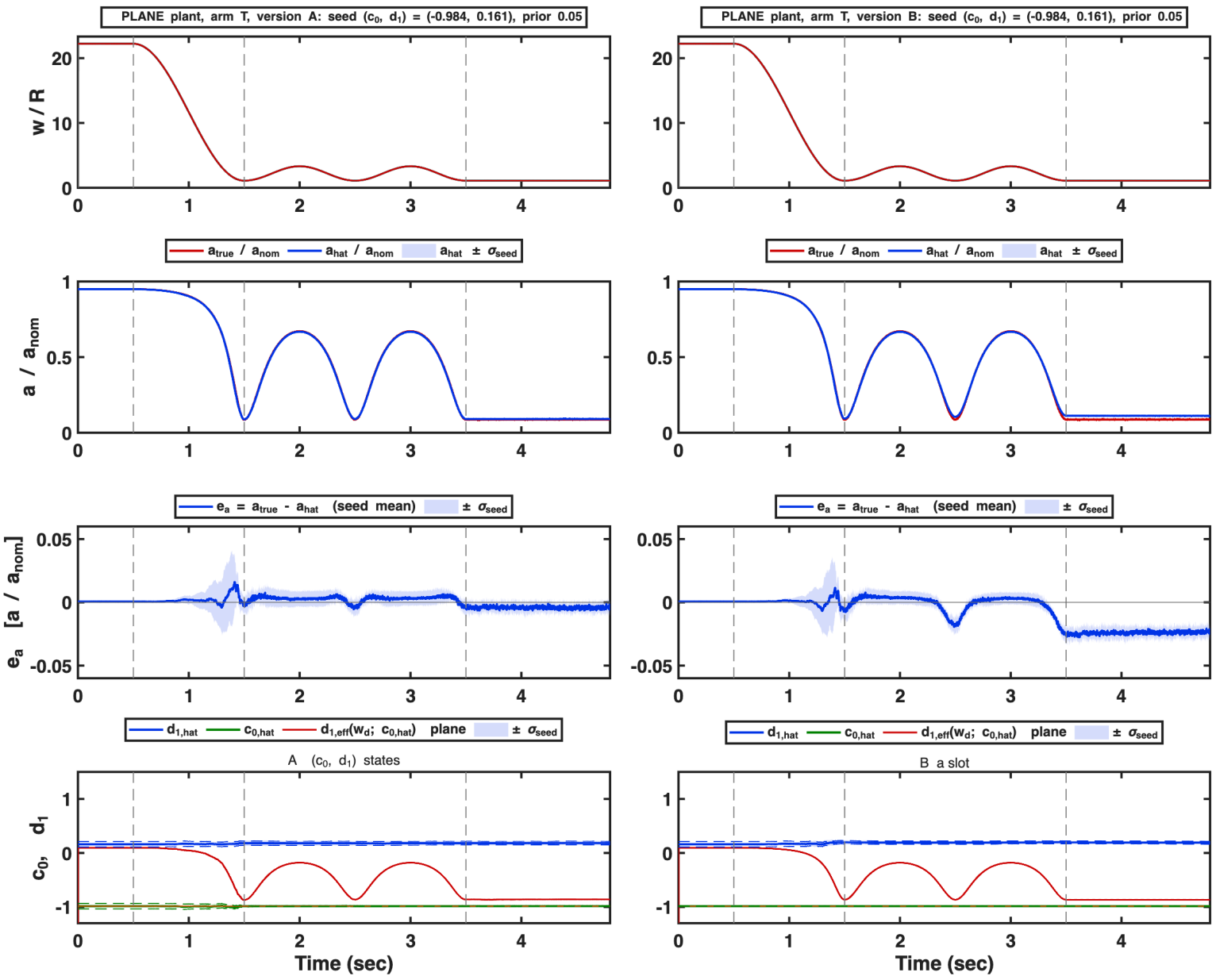

canon A descent $t_0$ = 1.25 s
$\delta w_1$ -0.00 -0.00 1.00 -0.00 -0.00
$\delta w_2$ -0.00 -1.00 0.00 -0.00 -0.00
$\delta w_3$ -1.00 -0.00 0.00 -0.02 -0.00
$c_0$ -0.01 -0.00 0.00 0.84 -0.54
$d_1$ 0.01 0.00 0.00 -0.54 -0.84
$v_1$ $v_2$ $v_3$ $v_4$ $v_5$
950 680 680 319 19
$\sigma_i$
canon B descent $t_0$ = 1.25 s
$\delta w_1$ 0.00 0.00 -0.00 1.00 -0.00
$\delta w_2$ 0.00 0.00 -1.00 0.00 -0.00
$\delta w_3$ -0.06 -1.00 -0.00 0.00 -0.01
$a_w$ -1.00 0.06 -0.00 0.00 0.08
$d_1$ -0.08 0.01 -0.00 0.00 -1.00
$v_1$ $v_2$ $v_3$ $v_4$ $v_5$
2500 949 680 680 63
$\sigma_i$
canon A hold $t_0$ = 4.49 s
$\delta w_1$ -0.00 -0.00 1.00 -0.00 -0.00
$\delta w_2$ -0.00 -1.00 0.00 -0.00 -0.00
$\delta w_3$ -1.00 -0.00 0.00 0.00 -0.00
$c_0$ 0.00 -0.00 0.00 0.89 -0.45
$d_1$ -0.00 0.00 0.00 -0.45 -0.89
$v_1$ $v_2$ $v_3$ $v_4$ $v_5$
952 680 680 156 0
$\sigma_i$
canon B hold $t_0$ = 4.49 s
$\delta w_1$ -0.00 -0.00 1.00 -0.00 0.00
$\delta w_2$ -0.00 -1.00 0.00 -0.00 0.00
$\delta w_3$ -1.00 -0.00 0.00 0.03 0.00
$a_w$ -0.03 -0.00 0.00 -1.00 0.00
$d_1$ -0.00 -0.00 0.00 -0.00 1.00
$v_1$ $v_2$ $v_3$ $v_4$ $v_5$
955 680 680 318 0
$\sigma_i$
meng A descent $t_0$ = 8.75 s
$\delta w_1$ -0.00 -0.00 1.00 -0.00 0.00
$\delta w_2$ -0.00 -1.00 0.00 -0.00 0.00
$\delta w_3$ -1.00 -0.00 0.00 -0.00 0.00
$c_0$ -0.00 -0.00 0.00 0.87 0.49
$d_1$ 0.00 0.00 0.00 -0.49 0.87
$v_1$ $v_2$ $v_3$ $v_4$ $v_5$
952 680 680 123 1
$\sigma_i$
meng B descent $t_0$ = 8.75 s
$\delta w_1$ -0.00 -0.00 1.00 -0.00 0.00
$\delta w_2$ -0.00 -1.00 0.00 -0.00 0.00
$\delta w_3$ -1.00 -0.00 0.00 -0.02 0.00
$a_w$ -0.02 -0.00 0.00 1.00 -0.01
$d_1$ -0.00 -0.00 0.00 0.01 1.00
$v_1$ $v_2$ $v_3$ $v_4$ $v_5$
953 680 680 298 2
$\sigma_i$
meng A hold $t_0$ = 12.19 s
$\delta w_1$ -0.00 -0.00 1.00 -0.00 -0.00
$\delta w_2$ -0.00 -1.00 0.00 -0.00 -0.00
$\delta w_3$ -1.00 -0.00 0.00 0.00 -0.00
$c_0$ 0.00 -0.00 0.00 0.89 -0.45
$d_1$ -0.00 0.00 0.00 -0.45 -0.89
$v_1$ $v_2$ $v_3$ $v_4$ $v_5$
952 680 680 154 0
$\sigma_i$
meng B hold $t_0$ = 12.19 s
$\delta w_1$ -0.00 -0.00 1.00 -0.00 0.00
$\delta w_2$ -0.00 -1.00 0.00 -0.00 0.00
$\delta w_3$ -1.00 -0.00 0.00 0.03 0.00
$a_w$ -0.03 -0.00 0.00 -1.00 0.00
$d_1$ -0.00 -0.00 0.00 -0.00 1.00
$v_1$ $v_2$ $v_3$ $v_4$ $v_5$
955 680 680 316 0
$\sigma_i$

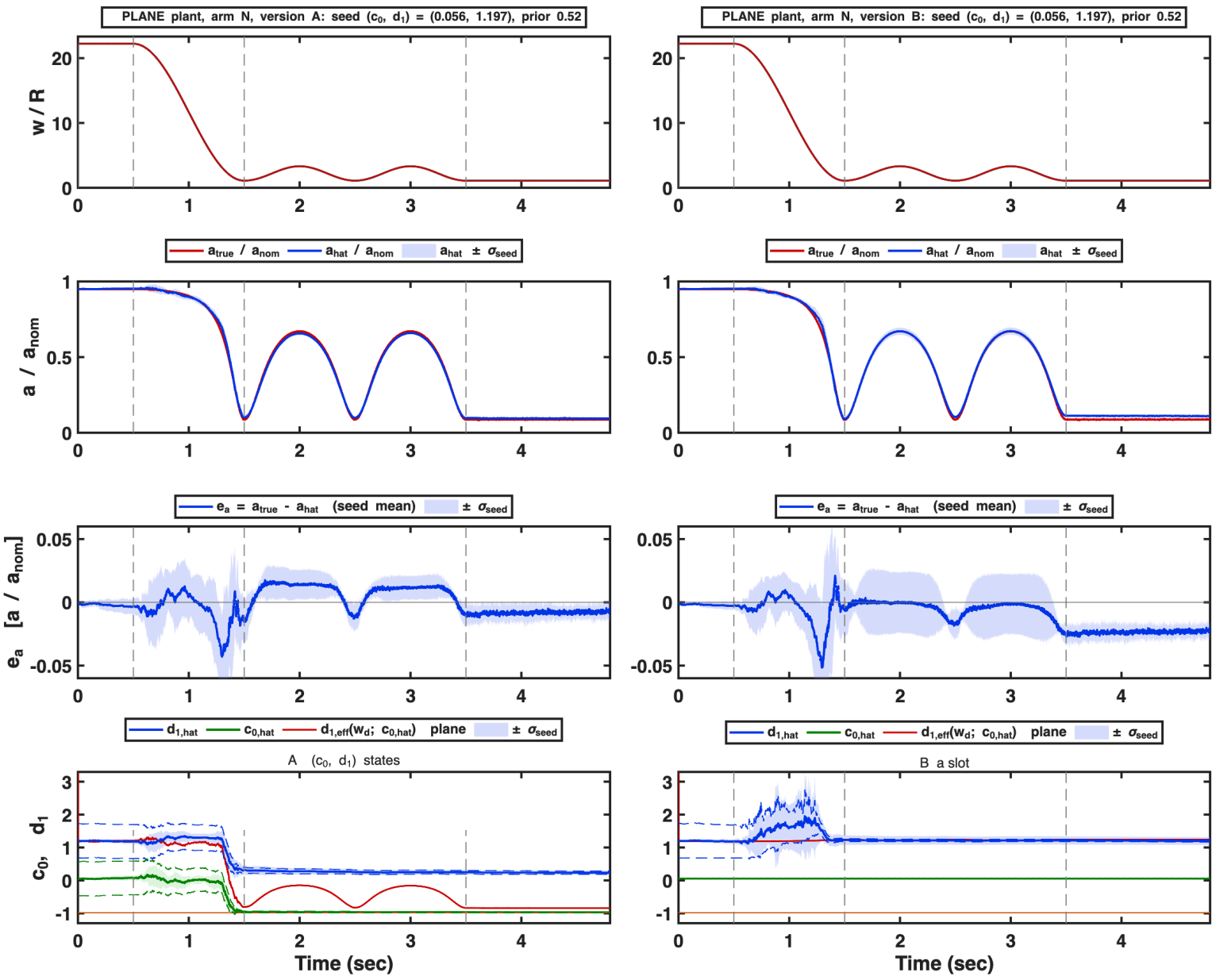

canon A descent $t_0$ = 1.25 s
$\delta w_1$ -0.00 0.00 -0.00 1.00 -0.00
$\delta w_2$ -0.00 0.00 -1.00 0.00 -0.00
$\delta w_3$ 0.08 1.00 -0.00 0.00 -0.01
$c_0$ 0.98 -0.08 -0.00 0.00 -0.19
$d_1$ -0.19 0.01 0.00 0.00 -0.98
$v_1$ $v_2$ $v_3$ $v_4$ $v_5$
1035 949 680 680 155
$\sigma_i$
canon B descent $t_0$ = 1.25 s
$\delta w_1$ 0.00 0.00 -0.00 1.00 -0.00
$\delta w_2$ 0.00 0.00 -1.00 0.00 -0.00
$\delta w_3$ -0.06 -1.00 -0.00 0.00 -0.01
$a_w$ -0.99 0.06 -0.00 0.00 0.16
$d_1$ -0.16 0.02 -0.00 0.00 -0.99
$v_1$ $v_2$ $v_3$ $v_4$ $v_5$
1.3e4 948 680 680 191
$\sigma_i$
canon A hold $t_0$ = 4.49 s
$\delta w_1$ -0.00 -0.00 -0.00 1.00 -0.00
$\delta w_2$ -0.00 -0.00 -1.00 0.00 -0.00
$\delta w_3$ -0.01 1.00 -0.00 0.00 -0.00
$c_0$ 1.00 0.01 -0.00 0.00 -0.08
$d_1$ -0.08 -0.00 0.00 0.00 -1.00
$v_1$ $v_2$ $v_3$ $v_4$ $v_5$
1567 959 680 680 4
$\sigma_i$
canon B hold $t_0$ = 4.49 s
$\delta w_1$ -0.00 -0.00 -0.00 1.00 -0.00
$\delta w_2$ -0.00 -0.00 -1.00 0.00 -0.00
$\delta w_3$ -0.66 -0.75 -0.00 0.00 -0.00
$a_w$ -0.75 0.66 -0.00 0.00 -0.00
$d_1$ 0.00 -0.00 0.00 0.00 -1.00
$v_1$ $v_2$ $v_3$ $v_4$ $v_5$
2430 746 680 680 27
$\sigma_i$
meng A descent $t_0$ = 8.75 s
$\delta w_1$ -0.00 -0.00 1.00 -0.00 0.00
$\delta w_2$ -0.00 -1.00 0.00 -0.00 0.00
$\delta w_3$ -1.00 -0.00 0.00 -0.02 0.00
$c_0$ -0.02 -0.00 0.00 0.96 0.27
$d_1$ 0.00 0.00 0.00 -0.27 0.96
$v_1$ $v_2$ $v_3$ $v_4$ $v_5$
945 680 680 432 7
$\sigma_i$
meng B descent $t_0$ = 8.75 s
$\delta w_1$ -0.00 -0.00 -0.00 1.00 0.00
$\delta w_2$ -0.00 -0.00 -1.00 0.00 0.00
$\delta w_3$ -0.87 -0.50 -0.00 0.00 0.00
$a_w$ -0.50 0.87 -0.00 0.00 -0.06
$d_1$ -0.03 0.05 -0.00 0.00 1.00
$v_1$ $v_2$ $v_3$ $v_4$ $v_5$
1093 711 680 680 23
$\sigma_i$
meng A hold $t_0$ = 12.19 s
$\delta w_1$ -0.00 -0.00 -0.00 1.00 -0.00
$\delta w_2$ -0.00 -0.00 -1.00 0.00 -0.00
$\delta w_3$ -0.00 1.00 -0.00 0.00 -0.00
$c_0$ 1.00 0.00 -0.00 0.00 -0.09
$d_1$ -0.09 -0.00 0.00 0.00 -1.00
$v_1$ $v_2$ $v_3$ $v_4$ $v_5$
1397 951 680 680 4
$\sigma_i$
meng B hold $t_0$ = 12.19 s
$\delta w_1$ -0.00 -0.00 -0.00 1.00 0.00
$\delta w_2$ -0.00 -0.00 -1.00 0.00 0.00
$\delta w_3$ -0.64 -0.76 -0.00 0.00 -0.00
$a_w$ -0.76 0.64 -0.00 0.00 -0.04
$d_1$ -0.03 0.03 -0.00 0.00 1.00
$v_1$ $v_2$ $v_3$ $v_4$ $v_5$
2488 750 680 680 34
$\sigma_i$

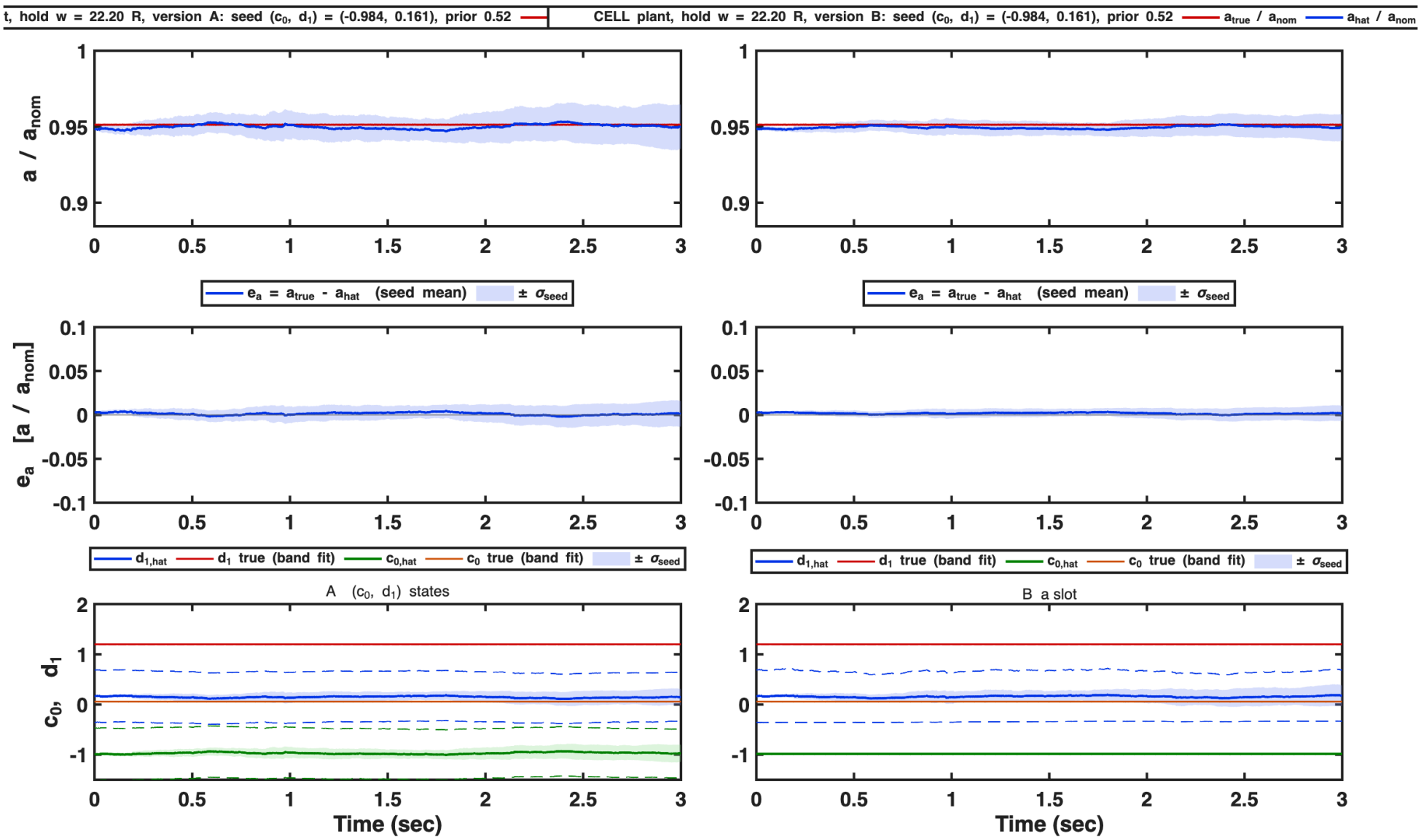

A: $c_0$ state
B: $a$ slot
canon, plane, descent
canon, plane, hold
canon, cell, descent
canon, cell, hold
meng, plane, descent
meng, plane, hold
meng, cell, descent
meng, cell, hold
$\sigma_i$
$i$

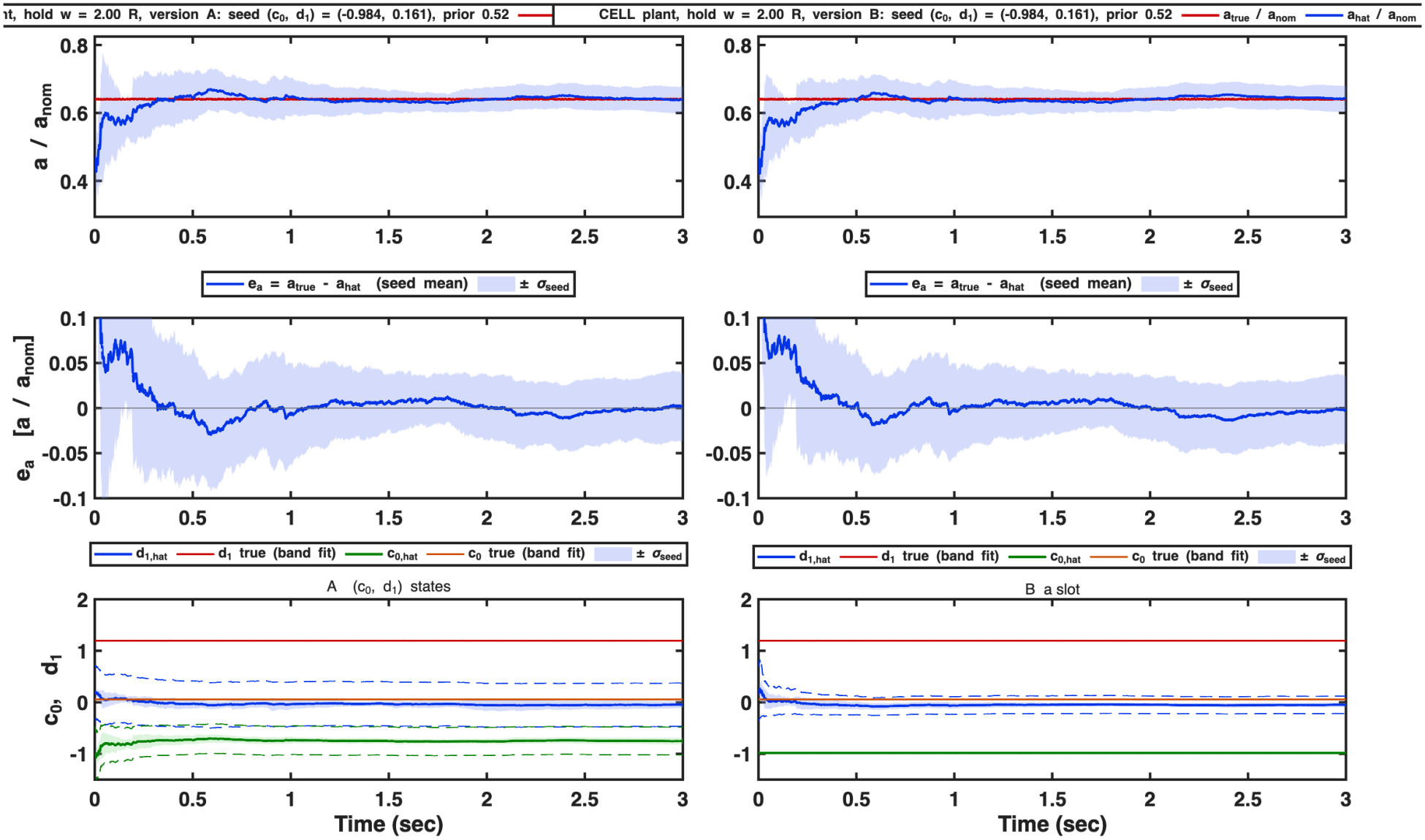

CELL plant, arm T, version A: seed (c0, d1) = (0.056, 1.197), prior 0.05
CELL plant, arm T, version B: seed (c0, d1) = (0.056, 1.197), prior 0.05
w / R
atrue / anom
ahat / anom
ahat ± σseed
a / anom
ea = atrue - ahat (seed mean) ± σseed
ea [a / anom]
d1,hat
c0,hat
d1 cell band fit ± σseed
A (c0, d1) states
B a slot
c0, d1
Time (sec)

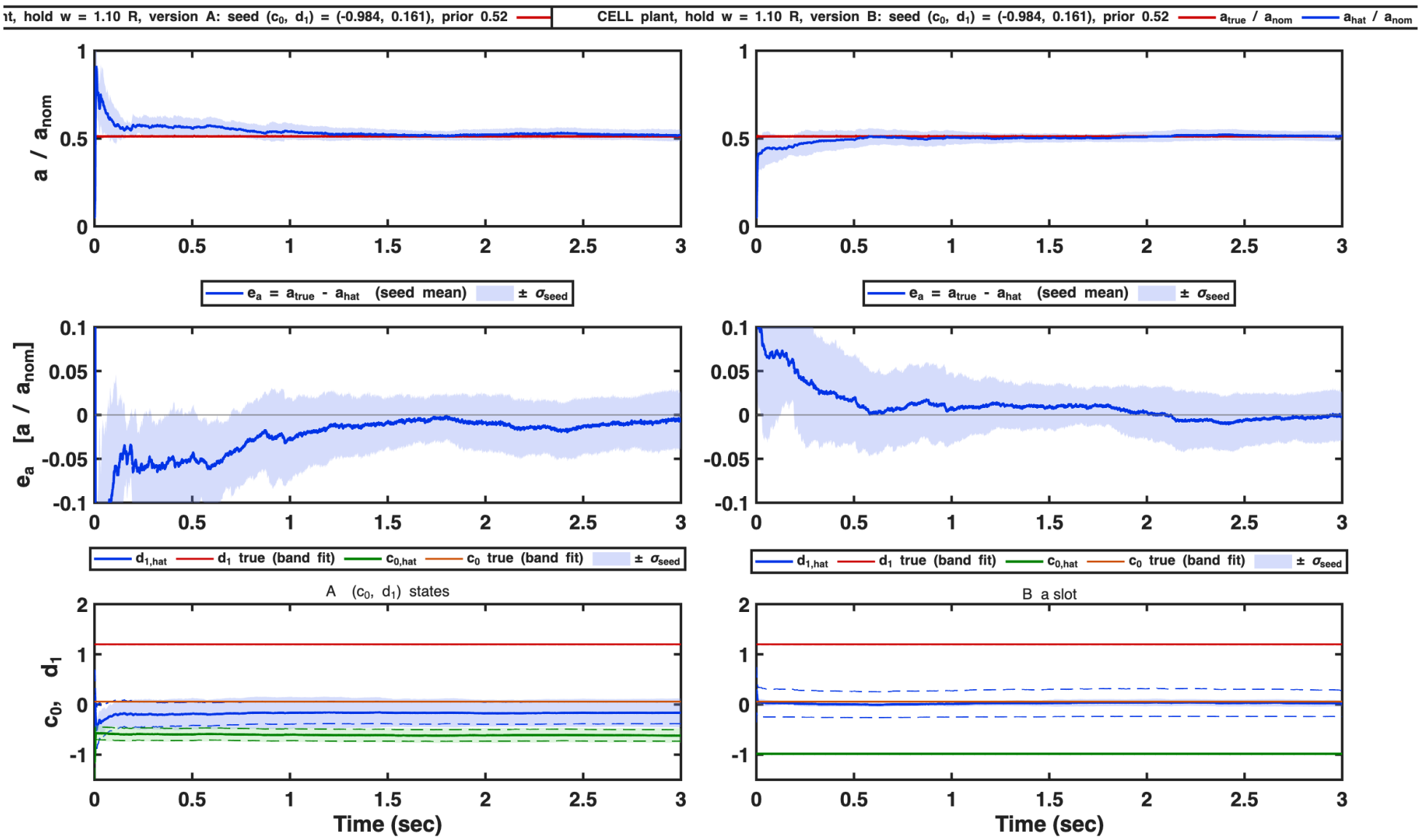

CELL plant, arm P, version A: seed (c0, d1) = (-0.984, 0.161), prior 0.52
CELL plant, arm P, version B: seed (c0, d1) = (-0.984, 0.161), prior 0.52
w / R
atrue / anom
ahat / anom
ahat ± σseed
a / anom
ea = atrue - ahat (seed mean) ± σseed
ea [a / anom]
d1,hat
c0,hat
d1 cell band fit ± σseed
A (c0, d1) states
B a slot
c0, d1
Time (sec)

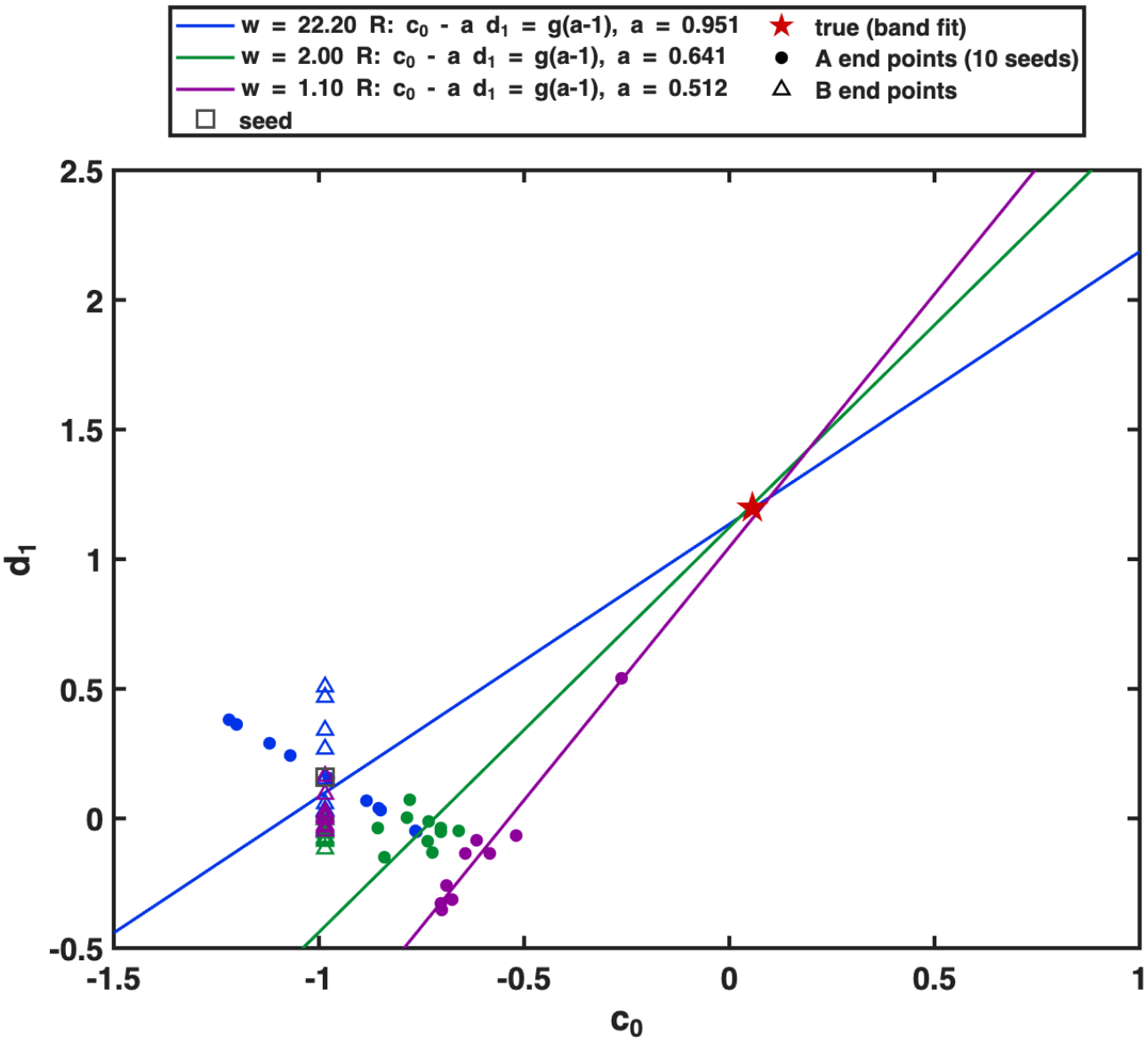

PLANE plant, arm T, version A: seed (c0, d1) = (-0.984, 0.161), prior 0.05
PLANE plant, arm T, version B: seed (c0, d1) = (-0.984, 0.161), prior 0.05
w / R
atrue / anom
ahat / anom
ahat ± σseed
a / anom
ea = atrue - ahat (seed mean) ± σseed
ea [a / anom]
d1,hat
c0,hat
d1,eff(wd; c0,hat) plane ± σseed
A (c0, d1) states
B a slot
c0, d1
Time (sec)

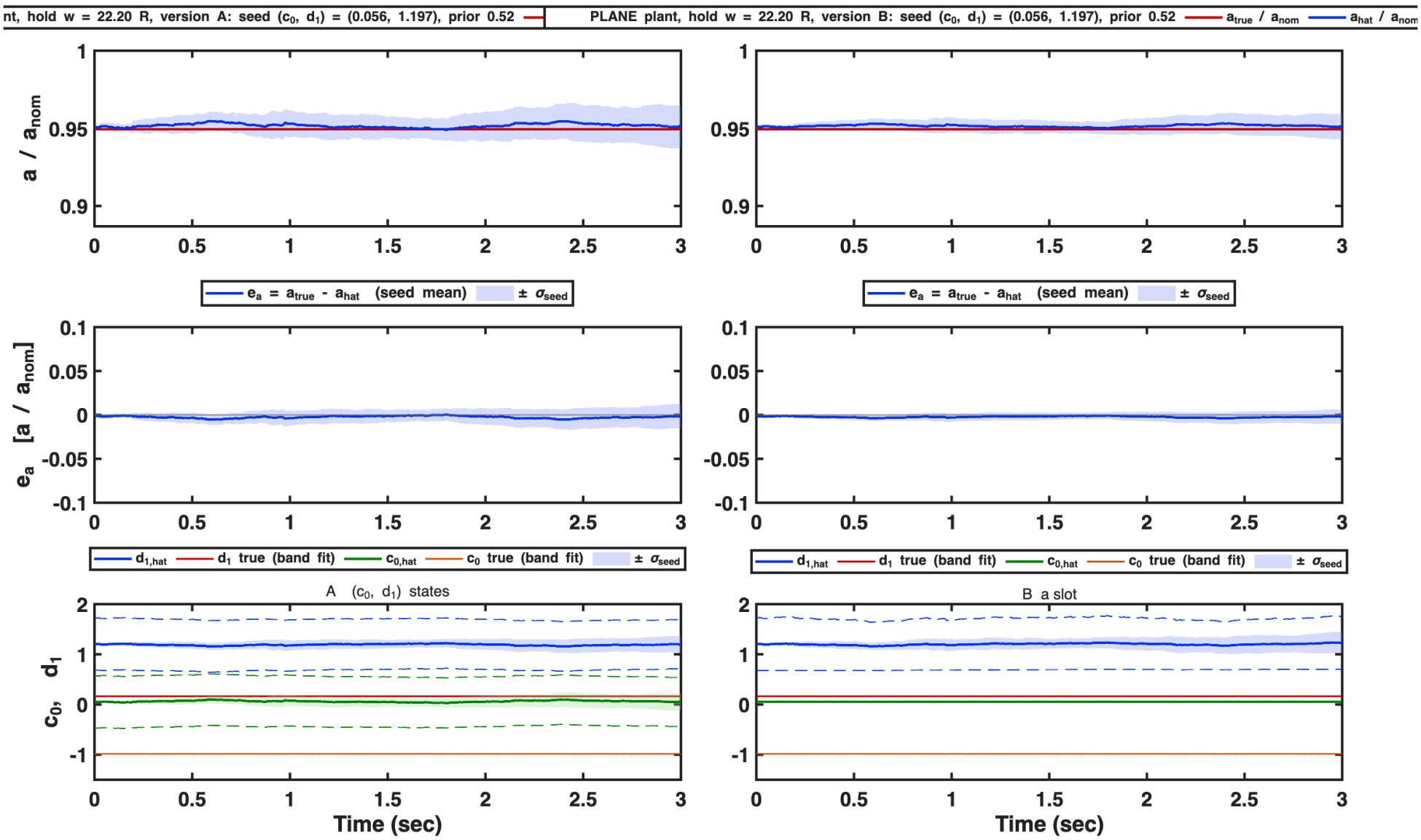

PLANE plant, arm N, version A: seed (c0, d1) = (0.056, 1.197), prior 0.52
PLANE plant, arm N, version B: seed (c0, d1) = (0.056, 1.197), prior 0.52
w / R
atrue / anom
ahat / anom
ahat ± σseed
a / anom
ea = atrue - ahat (seed mean) ± σseed
ea [a / anom]
d1,hat
c0,hat
d1,eff(wd; c0,hat) plane ± σseed
A (c0, d1) states
B a slot
c0, d1
Time (sec)

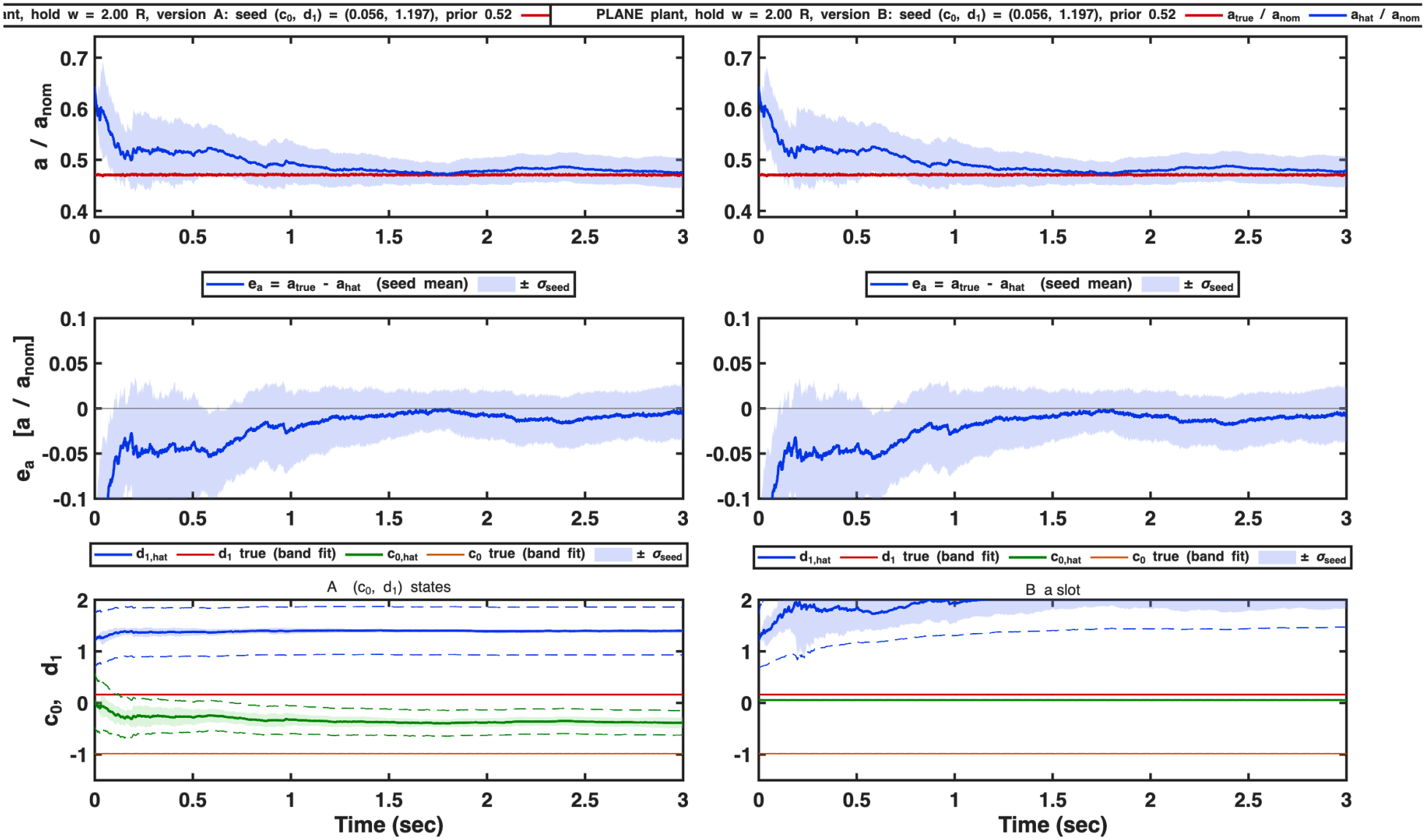

t, hold w = 22.20 R, version A: seed (c0, d1) = (-0.984, 0.161), prior 0.52
CELL plant, hold w = 22.20 R, version B: seed (c0, d1) = (-0.984, 0.161), prior 0.52
atrue / anom
ahat / anom
a / anom
ea = atrue - ahat (seed mean)
± σseed
ea [a / anom]
d1,hat
d1 true (band fit)
c0,hat
c0 true (band fit)
A (c0, d1) states
B a slot
c0, d1
Time (sec)

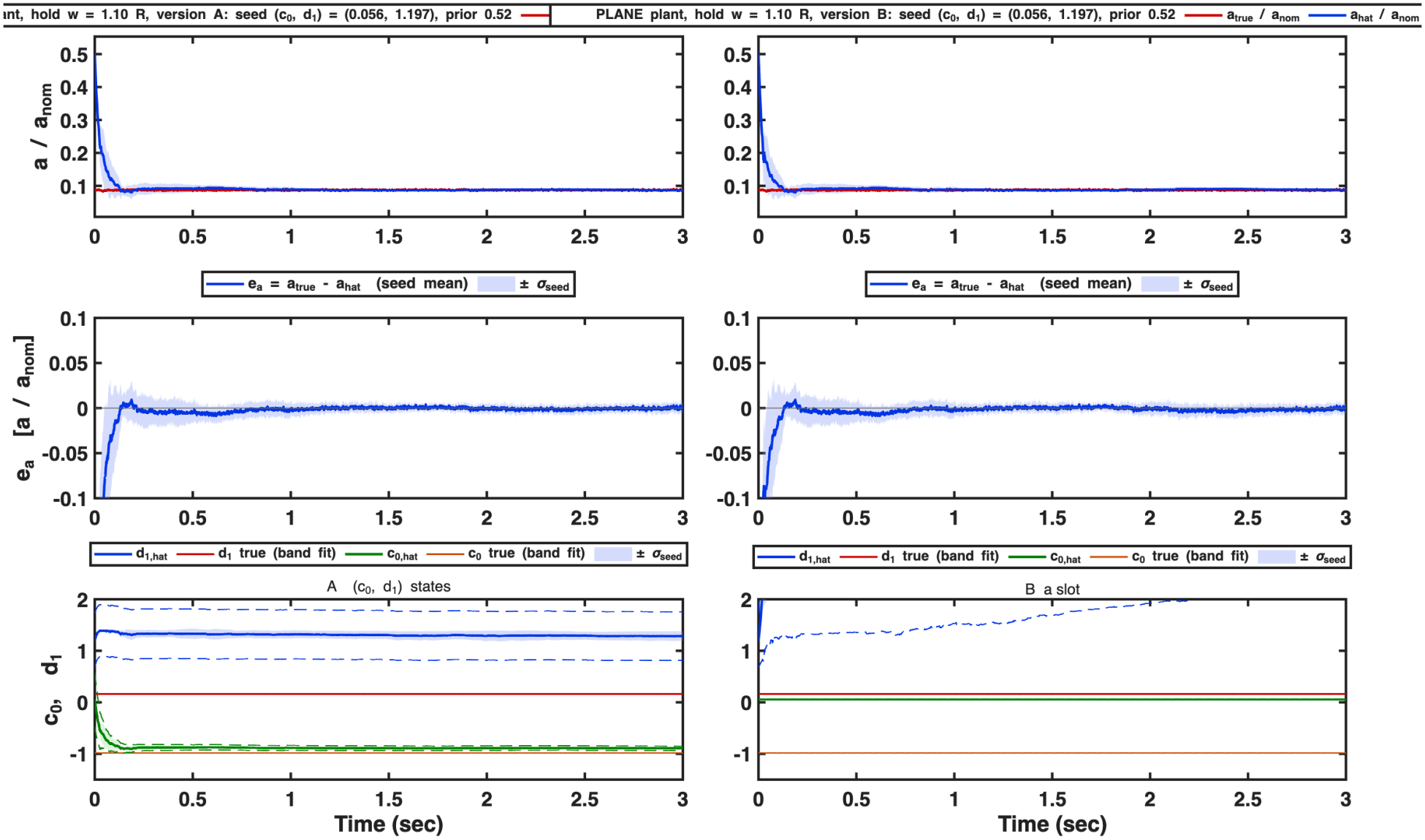

ıt, hold w = 2.00 R, version A: seed (c0, d1) = (-0.984, 0.161), prior 0.52
CELL plant, hold w = 2.00 R, version B: seed (c0, d1) = (-0.984, 0.161), prior 0.52
atrue / anom
ahat / anom
a / anom
ea = atrue - ahat (seed mean)
± σseed
ea [a / anom]
d1,hat
d1 true (band fit)
c0,hat
c0 true (band fit)
A (c0, d1) states
B a slot
c0, d1
Time (sec)

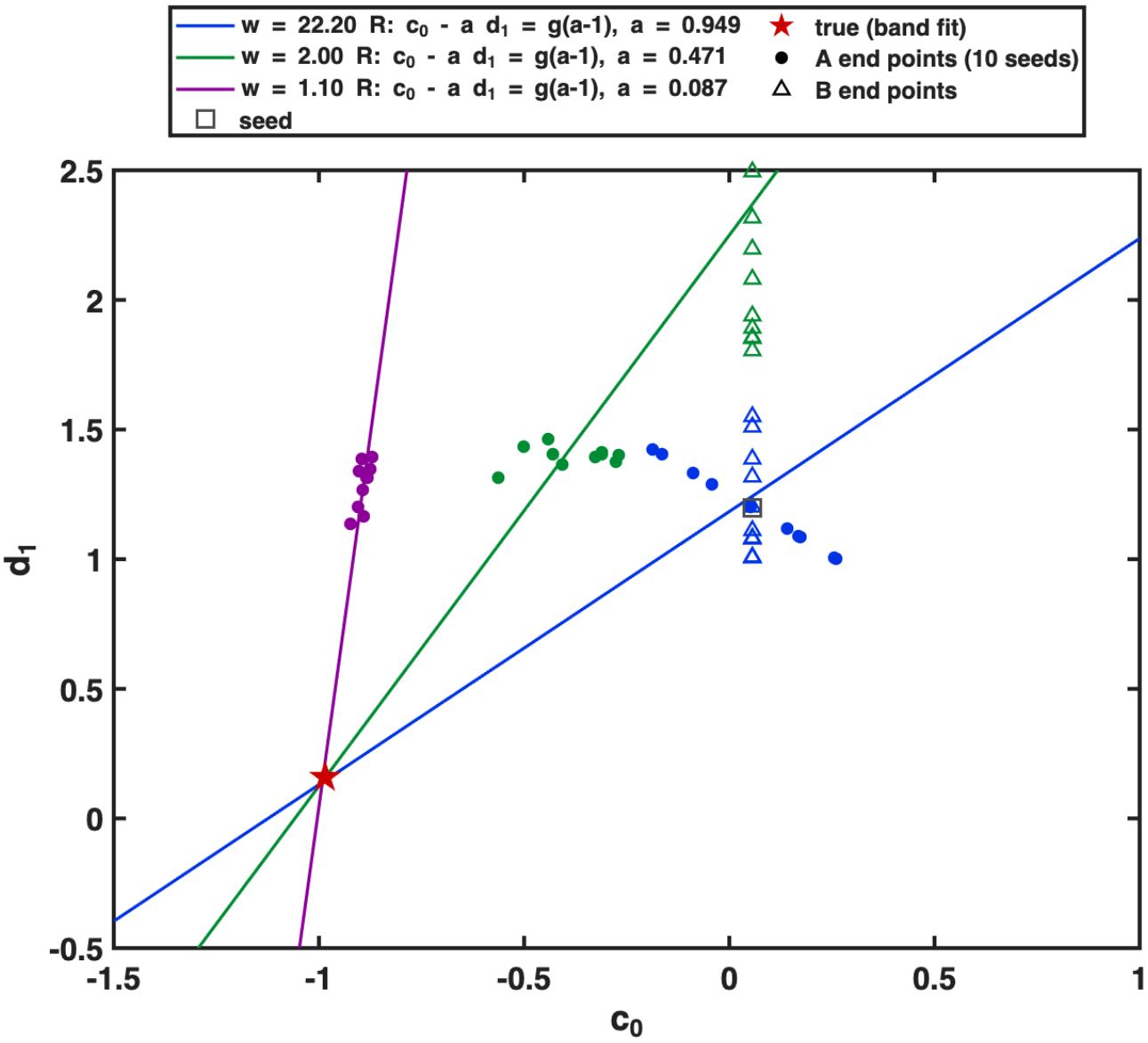

nt, hold w = 1.10 R, version A: seed (c0, d1) = (-0.984, 0.161), prior 0.52
CELL plant, hold w = 1.10 R, version B: seed (c0, d1) = (-0.984, 0.161), prior 0.52
atrue / anom
ahat / anom
a / anom
ea = atrue - ahat (seed mean)
± σseed
ea [a / anom]
d1,hat
d1 true (band fit)
c0,hat
c0 true (band fit)
A (c0, d1) states
B a slot
c0, d1
Time (sec)


w = 22.20 R: c0 - a d1 = g(a-1), a = 0.951
w = 2.00 R: c0 - a d1 = g(a-1), a = 0.641
w = 1.10 R: c0 - a d1 = g(a-1), a = 0.512
seed
true (band fit)
A end points (10 seeds)
B end points
2.5
2
1.5
1
0.5
0
-0.5
-1.5
-1
-0.5
0
0.5
1
d1
c0


nt, hold w = 22.20 R, version A: seed (c0, d1) = (0.056, 1.197), prior 0.52
PLANE plant, hold w = 22.20 R, version B: seed (c0, d1) = (0.056, 1.197), prior 0.52
atrue / anom
ahat / anom
a / anom
ea = atrue - ahat (seed mean)
± σseed
ea [a / anom]
d1,hat
d1 true (band fit)
c0,hat
c0 true (band fit)
A (c0, d1) states
B a slot
c0, d1
Time (sec)


ınt, hold w = 2.00 R, version A: seed (c0, d1) = (0.056, 1.197), prior 0.52
PLANE plant, hold w = 2.00 R, version B: seed (c0, d1) = (0.056, 1.197), prior 0.52
atrue / anom
ahat / anom
a / anom
ea = atrue - ahat (seed mean)
± σseed
ea [a / anom]
d1,hat
d1 true (band fit)
c0,hat
c0 true (band fit)
A (c0, d1) states
B a slot
c0, d1
Time (sec)


ant, hold w = 1.10 R, version A: seed (c0, d1) = (0.056, 1.197), prior 0.52
PLANE plant, hold w = 1.10 R, version B: seed (c0, d1) = (0.056, 1.197), prior 0.52
atrue / anom
ahat / anom
a / anom
ea = atrue - ahat (seed mean)
± σseed
ea [a / anom]
d1,hat
d1 true (band fit)
c0,hat
c0 true (band fit)
A (c0, d1) states
B a slot
c0, d1
Time (sec)


w = 22.20 R: c0 - a d1 = g(a-1), a = 0.949
w = 2.00 R: c0 - a d1 = g(a-1), a = 0.471
w = 1.10 R: c0 - a d1 = g(a-1), a = 0.087
seed
true (band fit)
A end points (10 seeds)
B end points
2.5
2
1.5
1
0.5
0
-0.5
-1.5
-1
-0.5
0
0.5
1
d1
c0